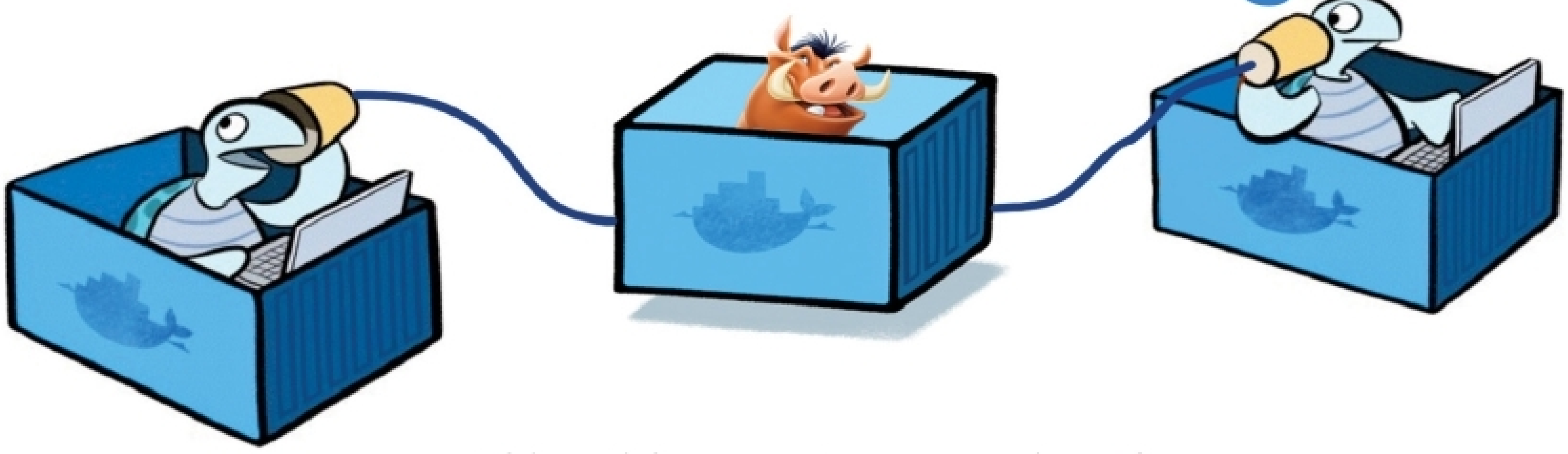
The Intricacies *of* Docker Networking



This article presents an overview of Docker networks and the associated networking concepts.

Customisation and the manipulation of default settings is an important part of software engineering. Docker networking has evolved from just a few limited networks to customisable ones. And with the introduction of Docker Swarm and overlay networks, it has become very easy to deploy and connect multiple services running on Docker containers – irrespective of whether they are running on a single or multiple hosts. To play around with Docker networking and tune it as per our requirements, we need to first understand the fundamentals as well as its intricacies. Assigning specific IP addresses to containers rather than using the default IPs, creating user-defined networks, and communication between containers running on multiple hosts are some of the challenges we will address in this article.

The idea of running machines on a machine is quite fascinating. This fascination was converted into reality when virtualisation came into existence. Virtual machines (VMs) are the basis of this process of virtualisation. VMs are being used everywhere—from small organisations to cloud data centres. They enable us to run multiple operating systems simultaneously with the help of hardware virtualisation.

Along with the IT industry's shift towards microservices (the small independent services required to support complete software suites or systems), a need arose for machines that consume less computing resources in comparison to VMs. One of the popular technologies that addresses this requirement is containers. These are lightweight in terms of resources, compared to VMs. Besides, containers take less time to create, delete, start or stop, when compared to VMs.

Docker is an open source tool that helps in the management of containers. It was launched in 2013 by a company called dotCloud. It is written in Go and uses Linux kernel features like namespaces and cgroups. It has been just five years since Docker was launched, yet, communities have already shifted to it from VMs.

When we create and run a container, Docker, by itself, assigns an IP address to it, by default. Most times, it is required to create and deploy Docker networks as per our needs. So, Docker lets us design the network as per our requirements.

Let us look at the various ways of creating and using the Docker network.

Aspects of the Docker network

There are three types of Docker networks—default networks, user-defined networks and overlay networks. Let us now discuss tasks like: (i) the allocation of specific IP addresses